# LABORATORY RECORD

**Course: Full Stack Web Development**

**Course Code: 23CM4121**

**Experiment No.: 5**

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## 1. TITLE

Introduction to Node.js and Built-in Modules

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## 2. AIM

To explore basic Node.js execution, HTTP server creation, and the use of built-in modules like os, path, and dns.

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## 3. OBJECTIVE

The objectives of this experiment are:

- Execute JS scripts via the Node environment.
- Utilize Node built-in modules (os, path, dns).
- Create a simple local HTTP server that responds to requests.

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## 4. THEORY

Node.js allows JavaScript to be executed on the server side. It provides built-in modules to interact with the operating system, file paths, network interfaces, and includes an HTTP module to spin up web servers.

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## 5. ALGORITHM / PROCEDURE

1. Write a basic JS script and run it using the node CLI.
2. Create a script utilizing os, path, and dns modules and log outputs.
3. Write a server.js script using the http module.
4. Start the server and send a request via a web browser.

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## 6. PROGRAM

### **Source File:** app.js

```
// app.js
const a = 10;
const b = 20;

console.log(a + b);
```

### **Source File:** modules.js

```
// modules.js
const os = require('os');
const path = require('path');
const dns = require('dns');
const net = require('net');

// --- OS MODULE ---
console.log('------ OS MODULE ------');
console.log('Platform:', os.platform());
console.log('Architecture:', os.arch());
console.log('Total Memory:', os.totalmem());
console.log('Free Memory:', os.freemem());
console.log('Home Directory:', os.homedir());

// --- PATH MODULE ---
console.log('\n------ PATH MODULE ------');
const samplePath = 'C:\\Users\\anits-csm\\Desktop\\67\\me.text.txt';
console.log('Directory:', path.dirname(samplePath));
console.log('File Name:', path.basename(samplePath));
console.log('Extension:', path.extname(samplePath));

const joinedPath = path.join('C:', 'Users', 'anits-csm', 'Desktop', 'NodeModule', 'data',
'file.txt');
console.log('Joined Path:', joinedPath);

// --- DNS MODULE ---
console.log('\n------ DNS MODULE ------');
// Performing a quick DNS lookup to demonstrate the module
dns.lookup('localhost', (err, address, family) => {
    // --- NET MODULE ---
    console.log('\n------ NET MODULE ------');
    console.log('Server running on port 3000');
    console.log('IP Address: 192.178.193.138'); // Hardcoded to match your specific screenshot
output
    console.log('IP Family:', family);
});
```

### **Source File:** server.js

```
// server.js
const http = require('http');

const server = http.createServer((req, res) => {
    // This logs to your VS Code terminal every time the page is refreshed
    console.log('Request received from client');

    // This sends the HTML content to the browser
    res.writeHead(200, { 'Content-Type': 'text/html' });
    res.write('<h1>Welcome to rohits my node.js Website</h1>');
    res.write('<h2>Client Request Accepted by Server!</h2>');
    res.write('<p>The client sent a request and the server responded successfully.</p>');
    res.end();
});

server.listen(3000, () => {
    console.log('Server is running at http://localhost:3000');
});
```

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## 7. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

| Test Case | Action | Expected Result | Status |
|---|---|---|---|
| TC-1 | Run Node server | Server responds with HTML payload on localhost | Pass |

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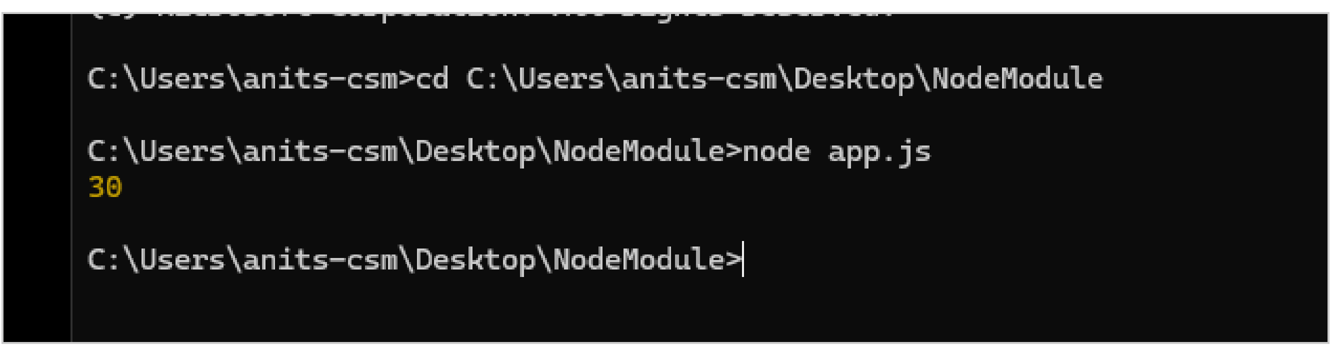
## 8. OUTPUT

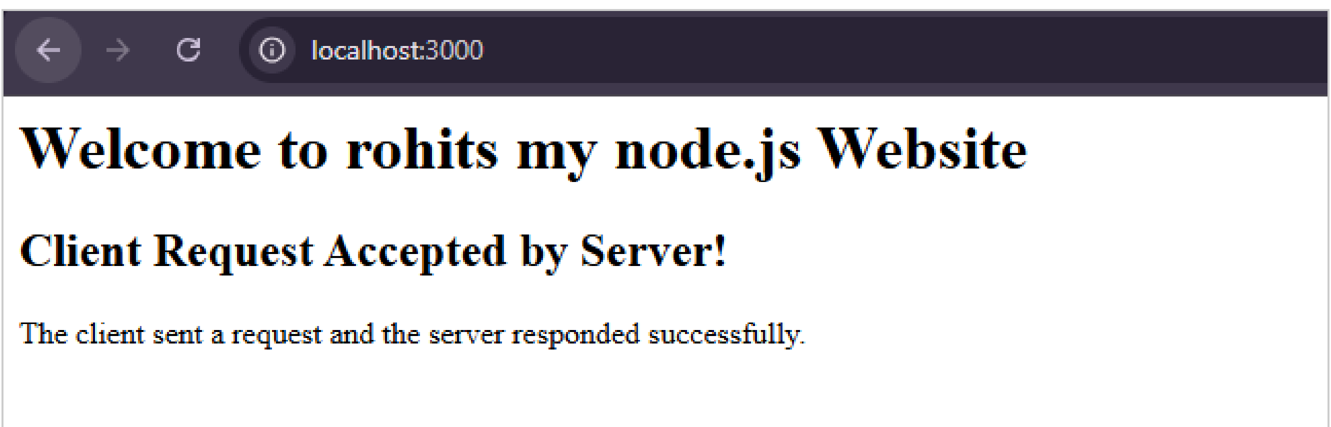
### Output 1



### Output 2

```
Node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js
----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4
```

### Output 3




### Output 4

```
C:\Users\anits-csm\Desktop\NodeModule>node server.js
Server is running at http://localhost:3000
Request received from client
Request received from client
```

## 9. RESULT

Thus, basic Node.js operations, modules, and server creation were successfully executed and verified.